

Research article

Integration of the SECI model and ChatGPT in higher education

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ARTICLE INFO

Keywords:

Knowledge management
Academic education
Artificial intelligence
Knowledge representation
Higher education

ABSTRACT

In educational settings, the socialization, externalization, combination, and internalization (SECI) model promotes collaboration, the creation of new knowledge, and the integration of individual and collective experiences, thereby enriching learning with greater depth and meaning. The objective of this study was to identify the factors that contribute to the optimal development of the SECI model through the use of ChatGPT, to understand their interaction. The experiment included undergraduate students of business sciences at public and private universities in Cusco, Huancayo, and Lima, Peru. The students were divided into control and experimental groups, with 100 students in each group. The main selection criteria were prior knowledge of and willingness to use ChatGPT and being enrolled in a finance course. The phases of the SECI model were evaluated using ChatGPT in the experimental group through group discussions, brainstorming sessions, the creation of a manual, and individual assessments, exploring the impact of ChatGPT usage on learning. A comparison and description of each phase was conducted. Results revealed that while the use of ChatGPT improved the quality of interaction and externalization of knowledge in the experimental group compared to the control group, it did not have a notable impact on the quality of combination or internalization of knowledge. A multinomial logistic regression analyzed the impact of ChatGPT usage in each phase. Findings showed that the SECI model without ChatGPT explained 63.4 % of the change in internalization, while with ChatGPT it was 41.3 %. A positive correlation was observed between the use of ChatGPT and the performance in objective tests during the externalization and combination phases. However, while it enhanced the generation of ideas in the early stages of learning, it did not necessarily increase tacit knowledge in the internalization phase. The importance of using tools such as ChatGPT carefully and selectively, especially in the context of acquiring specialized knowledge, was emphasized.

1. Introduction

Artificial intelligence (AI) is rapidly transforming higher education by presenting new opportunities to enrich learning and address persistent challenges in teaching and knowledge management. Tools such as ChatGPT have shown the potential to improve active participation and personalize learning by processing large amounts of data and learning from its interactions with users [1,2]. However, this disruptive capability also brings limitations, such as generating superficial responses and inconsistency in texts [2], raising concerns about academic integrity and the ability of the app to develop complex and critical skills in students [1–3].

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In this context, the socialization, externalization, combination, and internalization (SECI) model, widely used in knowledge management, offers a conceptual framework to analyze the potential and challenges of integrating AI into teaching and learning processes. This model describes four phases: socialization, externalization, combination, and internalization that allow for the conversion of tacit knowledge into explicit knowledge and vice versa [4]. In the educational context, the SECI model has been a valuable tool to promote knowledge creation and transfer, facilitate collaboration between students and teachers, and promote innovative educational practices [5,6]. Its application also improves the cocreation of knowledge and meaningful learning in the classroom [7].

Integration of ChatGPT into the SECI model has the potential to transform the educational experience by augmenting each of its phases. ChatGPT can support the socialization phase by acting as a moderator in group discussions, assist in externalization by organizing and clarifying the ideas generated by students, and contribute to combination by processing information from various sources to help structure complex concepts [8–10]. However, its effectiveness is not without challenges. ChatGPT faces notable limitations in socialization, as it cannot capture nonverbal cues and cultural nuances, which are essential for effective tacit knowledge exchange and trust building [11]. In externalization, it is difficult for the app to interpret human subjectivities and translate them into explicit academic structures [12]. During combination, its ability to synthesize and establish meaningful relations between diverse concepts is limited. Finally, in internalization, while it facilitates access to and distribution of explicit knowledge, its ability to promote critical reflection and deep learning is limited [11].

The literature review reveals that the application of AI chatbots has focused primarily on information dissemination and response automation [13], which leaves a gap in their capacity to support the complete cycle of knowledge creation and transfer in the SECI model. In addition, the rapid evolution of conversational AI raises debates about its role in learning personalization, pedagogical design, and academic integrity [14,15]. The gap between the current capabilities of AI and the pedagogical needs of higher education remains a critical area that requires further exploration.

This study addresses a gap in the literature by exploring how ChatGPT can be integrated into the SECI model, specifically in teaching financial competencies in higher education. Since these competencies require practical skills and critical analysis, ChatGPT could have a considerable impact on the learning of complex concepts and their practical applications. This study evaluates how ChatGPT facilitates each phase of the SECI model, acting as a moderator, idea facilitator, organizational assistant, and support for knowledge integration. This study is pioneering in investigating how ChatGPT can be integrated into each phase of the SECI model, an approach that has been little explored in previous research on AI and education. Unlike general studies on the use of AI, this research provides a detailed empirical analysis of the specific contributions of ChatGPT to the socialization, externalization, and combination of knowledge, while observing the results obtained in the internalization phase and evaluating its potential and limitations for enhancing learning processes in higher education.

2. Literature review

The integration of AI tools such as ChatGPT into education has generated notable interest owing to its potential to transform traditional teaching models. The SECI model, which encompasses socialization, externalization, combination, and internalization of knowledge, serves as a framework to analyze how AI can influence the transfer and creation of knowledge in classrooms. This review explores the application of ChatGPT in each phase of the SECI model, highlighting its strengths and limitations.

Socialization Phase: In the socialization phase of the SECI model, individuals share tacit knowledge through interaction and experience [4]. ChatGPT facilitates and moderates group discussions, generating enthusiasm among students for its innovative features and adaptability [16,17]. It can guide debates, pose questions, and facilitate reflection and exchange of relevant experiences. However, ChatGPT has limitations. It does not capture nonverbal cues such as body language or tone of voice, which are key elements for building trust and strengthening relations in tacit knowledge exchange [18]. Moreover, it may struggle to grasp cultural nuances, limiting its ability to cultivate deep understanding among students. While it facilitates dialogue, it cannot replace those human interactions necessary for collaborative learning [11].

Externalization Phase: In the externalization phase, tacit knowledge is transformed into explicit knowledge [4]. ChatGPT is particularly useful at this stage, helping students to organize and structure their ideas. Its ability to generate real-time text and provide clear examples facilitates the expression of thoughts and the exchange of ideas [19]. In activities such as brainstorming and mind mapping, ChatGPT acts as a facilitator, transforming abstract ideas into more concrete and coherent structures [20]. However, concerns have been raised about potential dependency. By offering immediate and organized responses, students may rely on ChatGPT instead of developing their own critical thinking processes [21]. This could hinder the development of essential cognitive skills such as critical reflection and independent evaluation [22]. Moreover, ChatGPT does not always interpret human subjectivities accurately, which can reduce its effectiveness in translating complex or emotionally charged thoughts into coherent academic structures [23].

Combination Phase: In the combination phase, explicit knowledge is organized to form a broader body of knowledge [4]. ChatGPT can help students gather information from various sources and create coherent texts, which is particularly useful for complex tasks requiring the integration of multiple concepts [10]. However, ChatGPT faces challenges in areas requiring precision, such as mathematics and the exact sciences, where it has provided both correct and incorrect answers, affecting its reliability [23]. In accounting evaluations, students outperformed ChatGPT considerably [24]. This suggests that while ChatGPT helps organize information, it has limitations in synthesizing complex concepts and requires human oversight to avoid errors in knowledge consolidation.

Internalization Phase: In the internalization phase, students apply knowledge in their own context [4]. ChatGPT facilitates access to explicit information and helps to clarify concepts, which are useful in fields such as math education, where it assists with basic questions and problems [11,25]. However, ChatGPT has limitations in promoting deep learning and critical reflection. Although useful for memorization, it does not foster the development of metacognitive skills or critical analysis [25]. Furthermore, in areas such as

financial mathematics, ChatGPT has struggled to provide clear and accurate references, preventing the construction of new knowledge [26]. To improve precision in numerical contexts, alternatives such as ComputeGPT, which runs real-time code for responses, have been proposed [26].

3. Materials and methods

In this research, a “post-test” design was implemented, as shown in Fig. 1. This design distinguishes between two groups: the experimental group and the control group. The experimental group is exposed to the stimulus or treatment, which in this case is the use of ChatGPT, followed by a measurement to evaluate the results of this intervention. Meanwhile, the control group does not receive treatment but is also subjected to a measurement to compare the results without the influence of the stimulus. This design allows measurement and comparison of how the intervention (use of ChatGPT) affects the development of the socialization, externalization, and combination phases of the SECI model and subsequently evaluates the process of internalization of knowledge in both groups. The structure of the “post-test” ensures that the intact groups can be assessed at the end of the experiment, allowing for the isolation of the specific effect of the intervention.

The research comprised 200 undergraduate students of management from various regions of Peru: 80 from a public university in Cusco, 50 from a private university in Huancayo, and 70 from a private university in Lima. The selection of participants was based on their prior knowledge on ChatGPT, interest in its use, and enrollment in a finance course. All students had previously completed courses related to the use of technology according to the curriculum of their programs. Regarding financial knowledge, an entry exam was administered at the beginning of the course, demonstrating a uniform level of understanding in this area. The students were divided into two groups of 100 each: a control group and an experimental group.

Fig. 2 illustrates the research framework, which began with a review of the literature to define the two main objectives. Subsequently, an experiment was conducted based on the four phases of the SECI model: socialization (through group discussions), externalization (through brainstorming and mind mapping), combination (creating a manual), and internalization (objective evaluation of acquired knowledge). This process allowed for the assessment of the impact of using ChatGPT on learning and knowledge application.

To validate the measurement instruments, a reliability analysis was performed using McDonald’s Omega technique, yielding values of 0.726 for the control group and 0.723 for the experimental group. Data were analyzed using two approaches: descriptive statistics to describe and compare differences between groups (Objective 1) and multinomial logistic regression analysis to explore the impact of the phases of the SECI model on learning (Objective 2). In this analysis, the phases of socialization, externalization, and combination were treated as independent variables, while internalization was the dependent variable. Finally, the findings were synthesized into contributions, limitations, and recommendations, providing a comprehensive conclusion to the research.

3.1. Use of ChatGPT

ChatGPT was chosen for its widespread adoption in the academic field and its ability to generate detailed and personalized responses, which facilitate learning and supports research tasks. Unlike other chatbots, it offers adaptive and fluid responses, allowing for more effective tutoring. Its accessibility and free availability make it especially useful for students. In the SECI model, the capabilities of ChatGPT are integrated into the phases of socialization, externalization, and combination. In the socialization phase, ChatGPT moderates discussions, simulating conversations that promote the exchange of experiences and tacit knowledge. During externalization, it guides brainstorming sessions and the creation of concept maps, helping to organize complex concepts and providing clear examples to effectively codify ideas. In combination, ChatGPT acts as a specialized assistant, collecting and systematizing information, allowing students to create detailed materials and structure content coherently, facilitating the consolidation of explicit knowledge. Version 3.5 selected for this study offers advantages such as rapid information retrieval and support in academic writing. However, it may have limitations in specialized critical analysis.

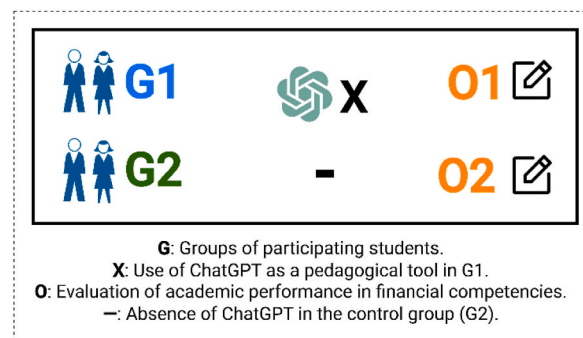


Fig. 1. The experimental design involves manipulating the presence or absence of the independent variable with the participation of an experimental group and a control group. An exclusive post-test design is used employing intact groups.

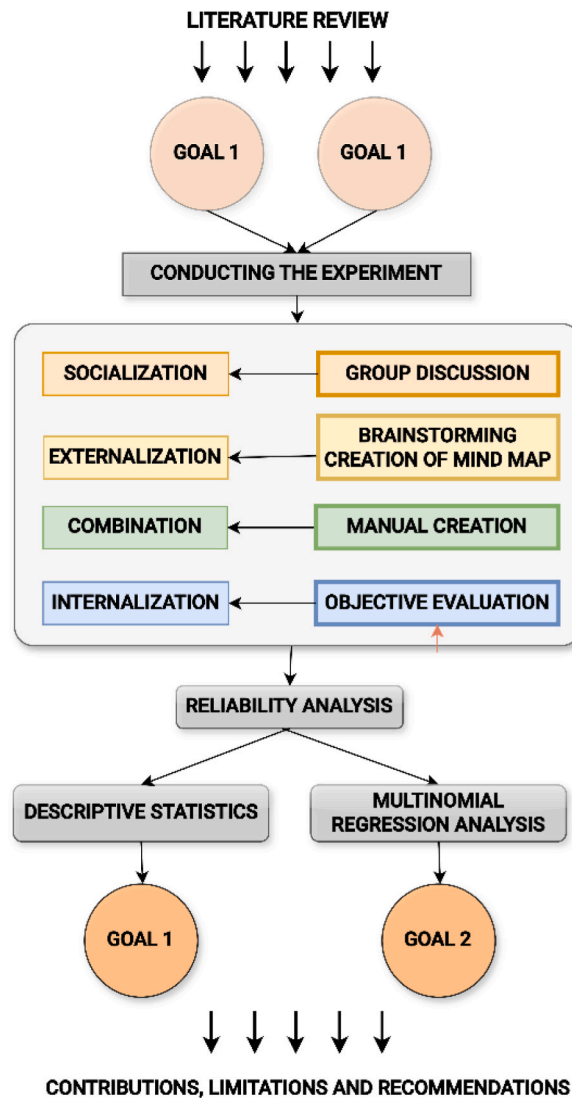


Fig. 2. Research framework.

3.2. Development of the experiment

For the development of the experiment, the activities and the main objectives of each phase of the SECI model were defined, as shown in Table 1. In these activities, participants discussed, organized, and applied financial concepts related to amortization. The phases were designed to facilitate the exchange, codification, consolidation, and practical evaluation of knowledge.

Table 1

Activities and Objectives in the Phases of the socialization, externalization, combination, and internalization (SECI) model in the Experiment.

Phase	Activity	Main Objective
Socialization	Group discussion on financial amortization (20 min). Groups of four students share their experience and knowledge.	Facilitate the exchange of tacit knowledge and improve the quality of group interaction.
Externalization	1) Brainstorming session (10 min) to generate concepts about financial amortization. 2) Creation of mind maps (20 min) to organize concepts visually, integrating ChatGPT.	Encourage the expression and codification of ideas and organize explicit knowledge in a structured manner.
Combination	Creation of a detailed manual on financial amortization (30 min), including basic concepts, calculations, and applied examples.	Consolidate and expand explicit knowledge, enhancing the ability to synthesize and apply financial concepts.
Internalization	Objective evaluation (40 min) through a written test with eight questions (theoretical and calculation based) on amortization.	Assess the degree of internalization and application of acquired financial knowledge in theoretical and practical contexts.

Table 2 highlights the differences between the control group and the experimental group, focusing on the integration of ChatGPT. This comparison indicates how the phases of the SECI model are developed: the control group follows a traditional, nontechnological approach, while the experimental group incorporates ChatGPT to enhance discussion, structure ideas, and create materials on financial amortization. The table specifically details how ChatGPT is integrated into each phase, optimizing the socialization, externalization, and combination of knowledge.

Fig. 3 illustrates the four phases of the SECI model (socialization, externalization, combination, and internalization) in the context of learning about financial amortization. Each phase is associated with a specific activity: an informal discussion (socialization), brainstorming and mind map creation (externalization), manual preparation (combination), and individual objective evaluation (internalization). The continuous flow of knowledge is highlighted, moving from interaction and dialogue to structuring and practical application, promoting active learning and the integration of explicit knowledge.

3.3. Control of variables and use of prompts

To ensure consistency in the use of ChatGPT and minimize the influence of external variables, such as differences in instructions, components were established for developing prompts for the experimental group. These components were carefully designed to guide interactions with ChatGPT during each phase of the process, providing context, role, and specific tasks for the chatbot, as detailed in **Table 3**. This approach ensures clarity, alignment with objectives, and uniformity in the interactions.

With the essential components that the prompts needed to include in place, participants carefully structured the prompts to align the context, assigned role, and specific tasks. For example, during the socialization phase, prompts were developed with the following content: "You are a group discussion moderator (Role) on financial amortization (Context). Generate open-ended and specific questions that address basic concepts, personal experiences, and practical applications to foster dialogue (Task)." Similarly, during the externalization phase, prompts like the following were implemented: "You are a knowledge transformation facilitator (Role) (Context). Help participants generate key ideas about financial amortization and organize structured concepts with clear and practical examples (Task)."

3.4. Evaluation criteria for the phases

Specific evaluation criteria were established for each phase of the SECI model, aimed at measuring student participation and interaction with both content and technology, from the transfer of tacit knowledge to the consolidation of explicit knowledge. **Table 4** summarizes these criteria, which reflect key aspects of the cognitive and social activities of the participants, including the use of ChatGPT. In the internalization phase, which is not included in the table, knowledge assimilation was assessed through an objective test comprising eight questions, graded on a Likert scale from 1 to 5.

4. Results and discussion

4.1. Results

Fig. 4 presents the results of the socialization phase, where the students exchanged knowledge about financial amortization. In the control group, most of the students fell into low to moderate levels of interaction, with 38 % in the "low" category and 40 % in "moderate," while only 22 % reached "good" or "excellent" levels. In contrast, the experimental group, which used ChatGPT, showed a more balanced distribution toward higher levels, with 52 % in the "moderate," 33 % in the "good," and 3 % in the "excellent" categories. These results indicate that the inclusion of ChatGPT in the socialization process improved the quality and level of student interaction compared with the control group.

Fig. 5 shows the results of the externalization phase, where students participated in brainstorming sessions and created mind maps on financial amortization. In the control group, 24 % of the students showed a low level of externalization, while 47 % reached a

Table 2

Comparison of activities between the control group and the experimental group in the phases of the SECI model.

Phase	Control Group	Experimental Group (with ChatGPT)
Socialization	Students engage in a free discussion, without the use of technology, based solely on verbal exchange.	ChatGPT is used to guide conversation, enrich the exchange of ideas, and facilitate reflection on experiences related to financial amortization.
Externalization	1) Ideas are generated verbally without technological tools. 2) Mind maps are created manually based on group discussions.	1) ChatGPT assists in the generation and organization of ideas during brainstorming. 2) ChatGPT helps create mind maps by helping to create visual structures and clarification of concepts.
Combination	Students create the manual without the support of ChatGPT, using only class materials and prior discussions.	ChatGPT is used to generate detailed and varied examples, verify mathematical calculations, and enrich the manual with FAQs and practical tips.
Internalization	The evaluation is conducted without the intervention of ChatGPT, measuring the knowledge acquired from previous activities.	Although ChatGPT is not used in this phase, the evaluation assesses the understanding and application of the knowledge supported by ChatGPT in earlier phases.

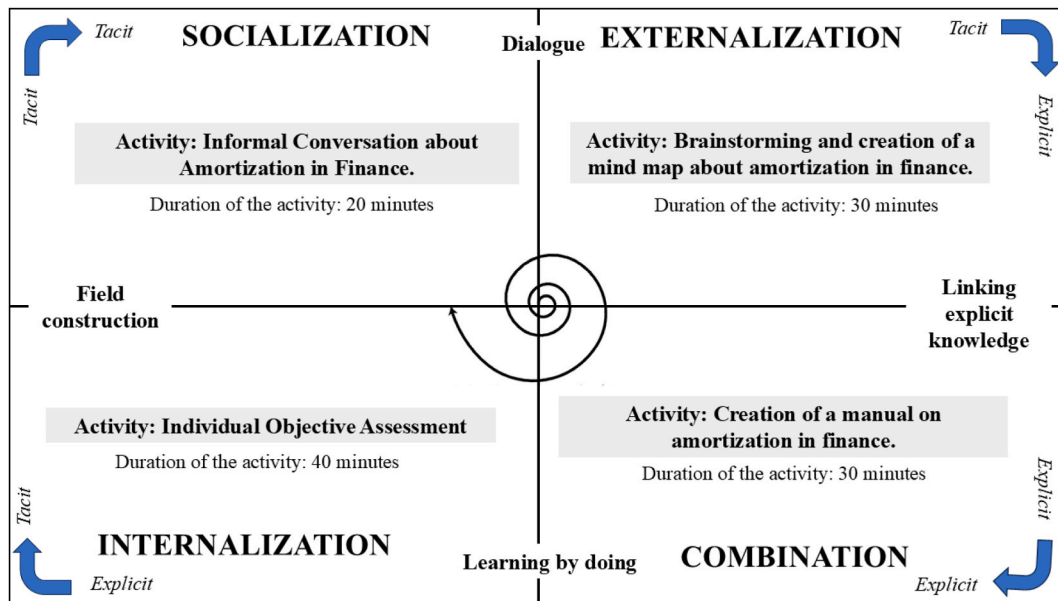


Fig. 3. Stages of the SECI model applied in the experiment.

Table 3

Components of prompts for the socialization, externalization, and combination phases.

Phase	Context	Role	Tasks
Socialization	Group discussion on financial amortization	Group discussion moderator	Generate questions to stimulate dialogue. Facilitate conversation among participants. Guide the exchange of experience and knowledge.
Externalization	Brainstorming and creation of conceptual maps on amortization	Facilitator of knowledge transformation	Guide the brainstorming session. Help organize and structure concepts. Provide clear examples to illustrate ideas.
Combination	Creation of a detailed manual on financial amortization	Specialized assistant in information gathering	Gather data from various sources. Summarize key points. Structure the information in a coherent and clear manner.

“good” level. Only 15 % achieved an “excellent” level. In contrast, the experimental group, which used ChatGPT, had no students in the “low” or “moderate” levels. Most students reached higher levels, with 36 % achieving a “good” level and 64 % an “excellent” level. These results suggest that the use of ChatGPT facilitated a higher quality of externalization, strengthening its potential to assist in the expression and codification of tacit knowledge during collaborative activities such as idea generation and mind map creation.

Fig. 6 shows the results of the combination phase, in which the students created a financial amortization manual. In the control group, 47 % of the students reached a “moderate” level and 52 % achieved a “good” level. The experimental group, which used ChatGPT, showed similar results, with 40 % at the “moderate” level and 47 % at the “good” level, reflecting an almost identical distribution to the control group. Both groups had minimal representation at the “low” and “excellent” levels. These results indicate that the use of ChatGPT did not have a notable impact on the quality of the combination phase, as the experimental and control groups similarly consolidated the explicit knowledge generated during externalization to create a comprehensive and organized body of knowledge.

Fig. 7 shows the results of the internalization phase, where an individual objective evaluation was carried out for both groups. In the control group, 57 % of the students reached a “moderate” level, while 43 % reached a “low” level. However, the experimental group, which used ChatGPT, presented a similar distribution, with 57 % at the “low” level and 35 % at the “moderate” level, with no students in the “good” or “excellent” levels. These results indicate that there were no notable differences in knowledge internalization between the control group and the experimental group. Both groups integrated the acquired knowledge as part of their skills and personal understanding, although the control group demonstrated a slight advantage in internalization quality compared with the experimental group.

Multinomial logistic regression was chosen to analyze the effect of ChatGPT owing to its ability to handle categorical variables and predict the probability of belonging to each level. This allows a comparison of how ChatGPT impacts the phases of the SECI model. The technique helps identify patterns and relations between the independent variables (socialization, externalization, and combination)

Table 4
Evaluation criteria for the socialization, externalization, and combination phases.

Phase	Evaluation Criterion	Description	Author
Socialization	Active participation	The participant actively and significantly contributes to the exchange of knowledge.	[27]
	Facilitation of dialogue	The participant helps to create an environment conducive to dialogue and interaction among the participants.	[28]
	Sharing relevant experiences	The participant shares experiences, values, beliefs, and personal emotions relevant to the topic.	[29]
	Building relationships	The participant builds meaningful relationships during the socialization process.	[27]
	Development of Collective Understanding	The participant contributes to the development of a collective understanding of knowledge within the group.	[27]
Externalization	Inclusive and nonthreatening environment	In the group, the participant appears confident in expressing their ideas freely.	[30]
	Recording of all relevant ideas	The participant takes notes and records all ideas related to the topic, regardless of their tangentiality.	[30–32]
	Activation of prior knowledge	The participant activates previous knowledge and skills to evaluate existing information and seek new information.	[30,33]
	Utilization of stimulating techniques	The participant uses techniques that encourage participation.	[30,34]
	Adaptability and versatility	The participant adapts to and participates naturally in the brainstorming technique.	[35,36]
	Central concept	The central concept is clear, relevant, and well-defined, guiding the development of the mind map effectively.	[37,38]
	Main topics and branching	The main topics and branches are defined clearly, organized hierarchically, and related effectively to the central concept.	[37]
	Keywords and minimalism	All branches are labeled with relevant and concise keywords that improve understanding of the mind map.	[39,40]
	Color coding and visual elements	Color coding and visual elements are used creatively and coherently, significantly enhancing the understanding and visual appeal of the mind map.	[41]
	Links and relationships	Clear and meaningful links are established between all branches, demonstrating a deep understanding of conceptual relations within the mind map.	[42]
	Clarity and organization	The manual is structured logically and the information is presented clearly.	[43]
	Accuracy and reliability	The information provided is accurate and reliable, supported by appropriate sources.	[43]
Combination	Relevance and pertinence	The content of the manual is relevant to the target audience.	[43]
	Usability and accessibility	The manual is easy to use and access, with an intuitive design, a clear index, and an appropriate visual presentation.	[43]
	Effectiveness in knowledge transmission	The manual conveys knowledge effectively, facilitating the reader's learning and understanding of the topic addressed.	[43]

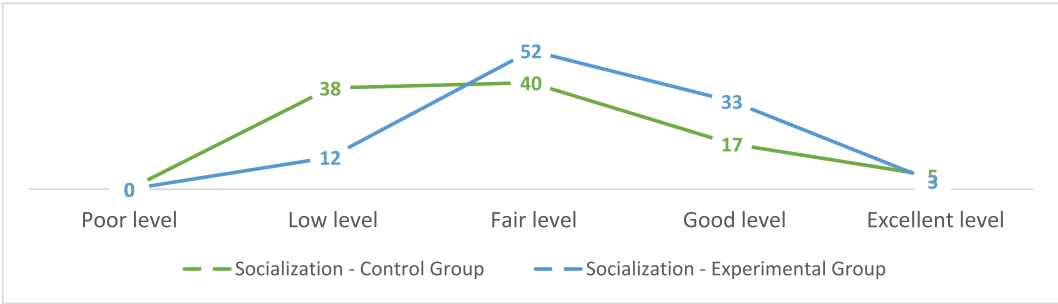


Fig. 4. Level of socialization.

and the dependent variable (internalization), providing a detailed analysis of the effect of ChatGPT on learning. Table 5 shows that the final model fits better than the initial beta value, supported by an AIC index of 70.436 and a significance level below 0.05, which confirms the fit of the multinomial regression model.

In Table 6, the values obtained in the pseudo-R-squared are shown, selecting the highest one, which is Nagelkerke's with a coefficient of 0.634. This indicates that the final model explains 63.4 % of the variance in the dependent variable (internalization), which is a notable value for measuring causality.

In Table 7, it can be seen that in the SECI model, the socialization factor does not predict the internalization score because its p-value of 0.916 is not statistically significant. However, according to the coefficient Exp (B), as socialization increases by one unit, the student is 1.032 times more likely to obtain an intermediate score in the objective test conducted for internalization.

The externalization factor predicts the score obtained in internalization, as its p-value of 0.000 is statistically significant. Furthermore, the coefficient of Exp (B) or the odds ratio indicates that as the externalization increases by one unit, the student is 3.488 times more likely to obtain an intermediate score in the objective test conducted for internalization. Similarly, the combination factor predicts the score obtained in internalization, since its p-value of 0.006 is statistically significant. However, the coefficient shows that as the combination increases by one unit, the student is 0.269 times more likely to obtain a basic score in the objective test conducted

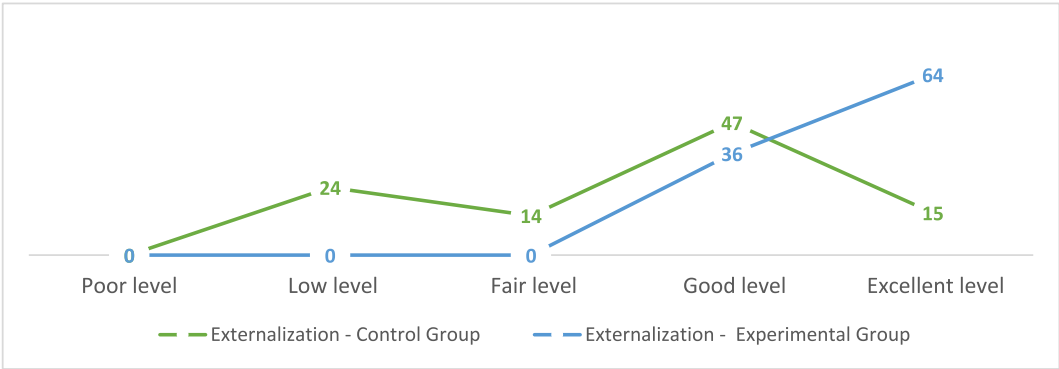


Fig. 5. Level of externalization.

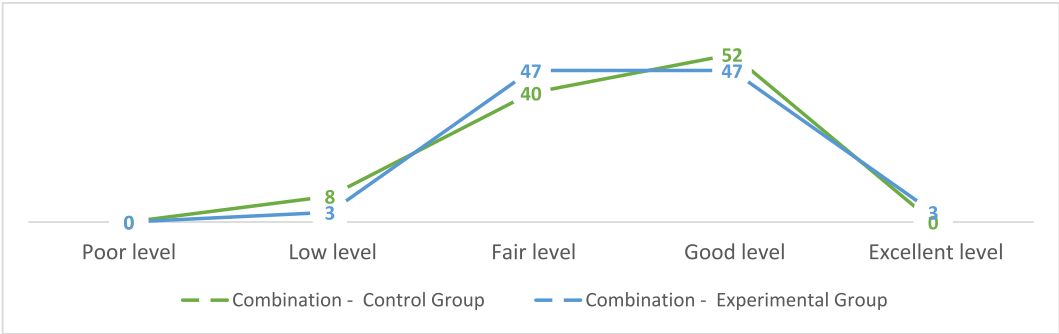


Fig. 6. Level of combination.

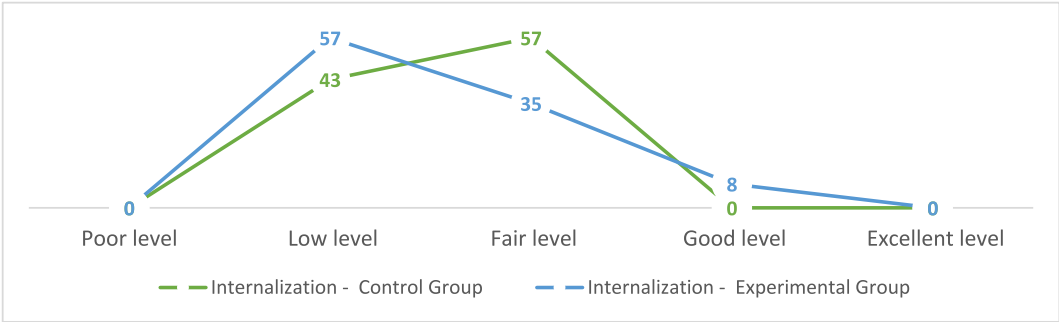


Fig. 7. Level of internalization.

Table 5
Model fit information–control group.

Model	Model Fit Criteria			Likelihood Ratio Tests		
	AIC	Normalized	Log Likelihood 2	Chi-square	df	Sig.
Only Intercept	118.48	121.553	116.948			
Final	70.436	93.883	52.436	64.512	8	<0.001

for internalization, which is the lowest of the three independent variables. From this point on, the results obtained in the experimental group are described using socialization, externalization, and combination as independent variables and internalization as a dependent variable. In Table 8, it is observed that the final model fits better than having only the beta value. This is demonstrated by the AIC index (96.713), which is lower compared to a significance of less than 0.05. Therefore, it complies with the multinomial regression model.

Table 6
Pseudo-R-squared–control group.

Cox y Snell	0.475
Nagelkerke	0.634
McFadden	0.465

Table 7
Parameter estimates–control group.

Internalization		B	Desv. Error	Std. Error	Wald	df	Sig.
Intermediate	Intercept	−0.021	1.379	0.000	1	0.988	
	Socialization	0.032	0.302	0.011	1	0.916	1.032
	Externalization	1.249	0.333	14.107	1	0.000	3.488
	Combination	−1.312	0.480	7.471	1	0.006	0.269

a. Reference category: Basic.

In [Table 9](#), the values obtained in the pseudo-R-squared are shown, selecting the highest, which is Nagelkerke's with a coefficient of 0.413. This indicates that the final model explains 41.3 % of the variance in the dependent variable (internalization), which is a considerable value for measuring causality.

In [Table 10](#), it can be seen that according to the results obtained, in the SECI model, the socialization factor predicts the internalization score because its p-value of 0.013 is statistically significant. Furthermore, the coefficient Exp (B) indicates that as socialization increases by one unit, the student is 0.156 times more likely to obtain a high score on the objective test conducted for internalization.

The externalization factor does not predict the internalization score owing to its p-value of 0.178, which is not statistically significant. Furthermore, the coefficient of Exp (B) or odds ratio indicates that as the externalization increases by one unit, the student is 3.330 times more likely to obtain a low score in the objective test conducted for internalization, as indicated by the negative beta coefficient.

Similarly, the combination factor also does not predict the internalization score, as its p-value of 0.84 is not statistically significant. However, the coefficient Exp (B) shows that as externalization increases by one unit, the student is 1.165 times more likely to obtain a high score in the objective test conducted for internalization.

Observing the results of the control and experimental groups, it is evident that the overall model does not improve with the incorporation of ChatGPT. This is demonstrated by comparing the pseudo-squared results with the SECI model without the use of ChatGPT, achieving 63.4 % predictability, while the SECI model with the use of ChatGPT achieves 41.3 % predictability. This implies that the use of AI does not contribute substantially to the increase of tacit knowledge in the internalization process.

However, a detailed analysis of each stage of the model reveals that in the externalization and combination phases, where the emphasis is on collecting theoretical information, the causal relation is positive with betas of 1.203 and 0.152, respectively, and significant in terms of the probability of improving the objective test with coefficients of 3.330 and 1.165.

5. Discussion

Research on the impact of ChatGPT within the SECI model reveals mixed results, highlighting the utility of AI in certain phases of the learning process, while also emphasizing its limitations in others. In the socialization phase, there was a notable improvement in interaction and student engagement in the experimental group, suggesting that ChatGPT facilitates idea generation and participation in collaborative discussions. The studies by Rahman and Watanobe [17] and Romero-Rodríguez et al. [9] support these findings, emphasizing that the innovative features of ChatGPT, such as personalized feedback and an accessible interface, motivate students to engage more actively in educational settings. However, despite these initial benefits, prolonged use of this tool raises questions about how dependence on it could affect students' critical thinking and creativity [22].

In the externalization phase, the experimental group demonstrated an improvement in the quality of the outputs generated, indicating that ChatGPT facilitates the management of tacit knowledge, which is often challenging to articulate in collaborative activities [20]. Its real-time text generation capability also contributed to the originality and quality of creative solutions [19]. However, concerns arise about the potential of overdependence on the tool, which could limit the development of critical skills essential for autonomous and creative learning.

In the combination phase, the impact of ChatGPT was less marked. No notable improvement was observed in knowledge consolidation compared with the control group, suggesting that AI still faces limitations in providing consistent answers in complex subjects such as mathematics and exact sciences. Li et al. [24] noted that while ChatGPT could generate readable and useful texts for the dissemination of financial information, its accuracy in technical areas such as financial mathematics was inconsistent. This highlights a critical limitation in using ChatGPT for tasks that require precision and a detailed understanding of numerical concepts, underscoring the need for other tools such as ComputeGPT to improve the accuracy in such activities. In the internalization phase, no considerable differences in knowledge retention were found between the experimental and control groups. Indeed, the control group showed a slight advantage in retaining new knowledge. This may be related to the lack of clear references provided by ChatGPT on

Table 8

Model fit information–experimental group.

Model	Model Fit Criteria			Likelihood Ratio Tests		
	AIC	Normalized AIC	Log Likelihood 2	Chi-square	df	Sig.
Only Intercept	109.378	114.589	105.378			
Final	96.713	138.395	64.713	40.666	14	0.000

Table 9

Pseudo-R-squared–experimental group.

Cox y Snell	0.344
Nagelkerke	0.413
McFadden	0.237

Table 10

Parameter estimates–experimental group.

Internalization a		B	Std. Error	Wald	df	Sig.	Exp (B)
High	Intercept	2.278	5.391	0.178	1	0.673	
	Socialization	−1.859	0.748	6.178	1	0.013	0.156
	Externalization	1.203	0.892	1.817	1	0.178	3.330
	Combination	0.152	0.767	0.039	1	0.842	1.165

a. Reference category: Basic.

specific topics such as financial mathematics, making it challenging to build new knowledge [25]. Despite this, the potential of ChatGPT as a pedagogical support tool for problem solving and lesson creation is acknowledged, although it still faces limitations in solving numerical problems.

According to this study, the overall impact of ChatGPT on the SECI model indicates that its contribution to the internalization process is limited, with a reduction in the predictability of the model from 63.4 % to 41.3 %. This suggests that, while ChatGPT facilitates phases such as socialization and externalization effectively, its use in the consolidation of tacit knowledge is less productive. This finding aligns with previous studies that have emphasized the need for human supervision and complementary pedagogical approaches to mitigate the potential negative consequences of AI dependency [26,44].

The research by Jensen et al. [15] presents an optimistic view of using ChatGPT in higher education, highlighting its ability to personalize and enhance learning assessments by integrating effectively into curricula as a complement to students' thinking and writing processes. While both studies acknowledge the educational utility of AI, this study takes a more cautious stance on its long-term impact on knowledge consolidation. Similarly, Saúde et al. [12] recognize the benefits of ChatGPT for academic performance and feedback; however, as in this study, they emphasize the need for pedagogical support to foster critical and ethical skills in students. Likewise, Damaševičius [13] and the results of this study indicate that although ChatGPT can enhance certain educational activities, its limitations in contextual understanding, personalization, and the development of complex cognitive skills account for its reduced impact in advanced learning stages. This reinforces the idea that while ChatGPT is a valuable complement in some phases of the educational process, it cannot replace the depth of human learning. However, the students reported high levels of satisfaction and progress in the lessons wherein it was used, suggesting a positive effect associated with the novelty of the tool.

The research by Leleparý et al. [10] highlights the use of ChatGPT in tasks such as translation and the development of reading skills through practical questions that promote critical thinking and collaboration. This approach was well received by students, who appreciated the interactive environment facilitated by AI. However, in this study, the limitations of ChatGPT in handling complex contexts and fostering deeper critical thinking are acknowledged. These findings emphasize that while useful in certain areas, its ability to support the internalization of knowledge is limited. Despite initial enthusiasm and its promotion of innovative practices, a cautious approach is needed to ensure the long-term reliability and effectiveness of ChatGPT as institutions address the challenges of its implementation.

The results of the study by Baidoo-Anu and Ansah [45] align with those of this study, highlighting the potential of ChatGPT to improve socialization and facilitate certain educational activities. However, both studies emphasize the limitations of the tool in terms of deep learning, particularly in its precision and personalization, which is also reflected in the findings of this study on its limited impact on the combination and internalization of knowledge. The main difference between the studies lies in the approach: Baidoo-Anu and Ansah examine the use of ChatGPT in the education system in general, while this study focuses on an empirical analysis within the SECI model, explaining the identification of moderate and specific impacts in certain phases of collaborative learning.

Meanwhile, a relevant aspect is highlighted in the research by Rudolph et al. [2], wherein the performance of several chatbots (including GPT-4, ChatGPT-3.5, Bing Chat, and Bard) in specific educational tasks was evaluated. While the usefulness of these chatbots is noted, their varying capabilities and limitations are also highlighted, making careful use essential in educational settings.

Similar to the findings of the present study, it is observed that the effectiveness of these tools is limited, whether in their ability to consolidate knowledge or due to their inconsistent performance and contextual errors. The divergence between both studies can be attributed to different approaches; while the present research examines the impact of ChatGPT throughout the entire learning process within the SECI model, the study by Rudolph et al. focuses on the quality of responses from various chatbots on specific higher education tasks.

In addition, Romero-Rodríguez et al. [9] present different findings, emphasizing the influence of habit and behavioral intention in the use of ChatGPT. While our study shows a greater impact in the early phases of learning with reduced effectiveness in later stages, Romero-Rodríguez et al. focus on usage behaviors.

Similarly, the research by Rezaev and Tregubova [1] aligns with the findings of this study, noting that ChatGPT enhances knowledge exchange and collaborative activities. However, while this study shows a limited impact in the advanced phases of the SECI model, such as combination and internalization, Rezaev and Tregubova emphasize the ability of ChatGPT to reduce task time, structure discussions more efficiently, and foster innovative solutions. These differences reflect variations in the depth of impact: while this work observes a moderate effect on deep learning, Rezaev and Tregubova highlight the practical use of ChatGPT in specific tasks and remote learning, suggesting that it enhances efficiency and creativity but has limited influence on knowledge consolidation. Likewise, Vicente-Yagüe-Jara et al. [46] support the findings of this study, indicating that ChatGPT has a positive impact on activities that require rapid idea generation, socialization, and problem solving. However, the divergence lies in the focus; this study centers on knowledge transfer and consolidation, whereas Vicente-Yagüe-Jara et al. evaluate creativity and its comparison between AI and humans. These differences may be attributed to the complexity of the tasks analyzed and the learning phases where AI is employed. Although ChatGPT facilitates certain processes, it continues to face challenges in tasks requiring deeper knowledge integration or higher levels of creativity.

The research by Qadhi et al. [11], although not directly assessing the impact of AI on collaborative learning or knowledge acquisition, provides a conceptual framework for its ethical implementation, emphasizing the importance of maintaining academic standards in higher education. Similarly, Akpan et al. [14] take a global perspective, analyzing how generative AI technologies are rapidly transforming education and research, but without exploring the quality of knowledge acquisition, as this study does with the SECI model. Finally, the SECI model and ChatGPT could be optimized with insights from Samsuryadi [6], who proposes how advanced technologies can improve curriculum design and better prepare students for the labor market, offering a structural approach to enhance the efficiency of the model.

6. Conclusions

The results revealed that although the use of ChatGPT improved the quality of interaction and externalization of knowledge in the experimental group compared to the control group, it had no notable impact on the quality of combination or internalization of knowledge. Furthermore, multinomial regression analysis indicated that the inclusion of ChatGPT in the SECI model did not improve its predictability. That is, the use of ChatGPT did not substantially enhance the SECI model, as evidenced by the decrease in the pseudo-R-squared in both groups (63.4 % without ChatGPT compared to 42.2 % with ChatGPT). Results indicate that ChatGPT can facilitate tacit knowledge management and enhance expression during collaborative activities; however, it has limitations in mathematical comprehension and the promotion of critical skills. Although AI is essential, difficulties in obtaining precise information and performing specialized numerical operations require careful consideration when implementing these technologies. The research also raises concerns about the reliability of the information provided by AI tools and the need for human supervision to ensure the quality of the content. Furthermore, it highlights the lack of a single tool to address all aspects of learning. Ultimately, the use of ChatGPT does not contribute substantially to tacit knowledge according to the SECI model, indicating that its use does not improve the predictability of the model. These findings underscore the need to understand the limitations and challenges when implementing technologies such as ChatGPT in educational settings. Valuable information is provided on the impact of ChatGPT on learning, highlighting the need for future research to explore its effective integration in different educational contexts and disciplines, and to better understand its limitations and potential.

CRediT authorship contribution statement

Urpi Barreto: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization.
Yasser Abarca: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization.

Data availability statement

The data will be available upon request.

Ethical approval

The study, being educational and non-invasive in nature, was conducted in full compliance with applicable ethical standards. Informed consent was obtained from all participants in accordance with established ethical principles and research regulations. This consent was secured through a transparent process that ensured respect for participants' rights, the anonymization of their data, and their entirely voluntary participation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study is part of the research efforts conducted at the Continental University and the National University of San Antonio Abad of Cusco. It stands as a testament to the dedicated efforts of both institutions to contribute to the advancement of knowledge and academic excellence across various fields.

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